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Project Name

Persons With Disabilities Bracelet (PWDB)

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بسم الله الرحمن الرحيم

الآية الكريمة

قال الله تعالى:

"يَرْفَعُ اللَّهُ الَّذِينَ آمَنُوا مِنْكُمْ وَالَّذِينَ أُوتُوا الْعِلْمَ دَرَجَاتٍ ۚ وَاللَّهُ بِمَا تَعْمَلُونَ خَبِيرٌ"

(سورة المجادلة: 11)

صدق الله العظيم

الإهداء

إلى (أمي وأبي)، اللذين كانا دائماً القوة التي استندت إليها في كل التحديات، حيث علمتاني أن الصبر هو مفتاح التغلب على الصعاب، وأن الدعاء هو الحماية من كل أزمة.

إلى (زملائي الأعزاء)، الذين ساندوني في كل خطوة، وكنتم رفقاء الدرب في مواجهة التحديات، فلولاً تعاوننا لما تجاوزنا كل عقبة.

إلى (أساتذتي) في جامعة العلوم والتقانة، الذين علمونا كيف نواجه الصعاب بالعلم، وكانوا مصدراً للإلهام في كل مرحلة، ومعا تجاوزنا الصعاب في أيام ثقال.

"إلى (دكتورة أغابي نبيل) مشرفتنا، التي كانت لنا النور في أوقات الظلام، والإلهام في لحظات الشك. شكراً لك على صبرك، وتوجيهاتك الحكيمة، التي كانت سبباً رئيسياً في وصولنا إلى ما نحن عليه اليوم.

"وما جزاء الإحسان إلا الإحسانُ

تظلُّ قلوبنا لكم أبداً على الدوام"

جزيل الشكر والتقدير لكم، ولن ننسى دعمكم أبداً."

شكر وعرفان

"بكل فخر وامتنان، أتوجه بجزيل الشكر والتقدير إلى أساتذتي الكرام،

مشرفي العزيز، وزملائي الأعزاء على دعمهم الكبير وتفانيهم المستمر في

هذه الظروف الصعبة التي يمر بها السودان. لقد كانت وقفنكم معنا مصدر

قوة وإلهام وسدا منيعاً في هذه الظروف وكان لتوجيهاتكم العميقة دور كبير

في تجاوز التحديات التي واجهناها. بفضل الله ثم بفضل جهودكم جميعاً،

وصلنا إلى ما نحن عليه اليوم. هذا الإنجاز هو ثمرة تعاوننا وجهودكم

المشتركة. أسأل الله أن يوفقنا جميعاً لمزيد من العطاء، وأن نكون دائماً خير

مثال على التضامن والعمل المشترك من أجل الوطن."

Abstract

Currently, there is a growing demand for providing support and assistance to individuals with special needs. Smart assistive technologies have become increasingly important and can significantly improve people's quality of life.

These technological solutions include wearable devices, environmental sensors, and communication and information technologies that help with continuous health monitoring and immediate detection of medical issues or emergencies.

This project aims to develop a smart electronic bracelet equipped with sensors to measure vital signs, along with a GPS module and SIM card, to link patient data to a dedicated website. The bracelet allows administrators to monitor the patient's condition in real-time, providing essential information such as location, health status, and emergency alerts.

It delivers alerts in critical situations and enables quick access to patient data by scanning a QR Code. This project seeks to enhance patient health monitoring, speed up emergency response, and improve remote healthcare services.

المستخلص

في الوقت الحالي، هناك طلب متزايد على توفير الدعم والمساعدة للأفراد ذوي الاحتياجات الخاصة. أصبحت التقنيات المساعدة الذكية أكثر أهمية ويمكن أن تساهم بشكل كبير في تحسين جودة حياة الأفراد.

تشمل هذه الحلول التكنولوجية الأجهزة القابلة للارتداء، وحساسات البيئة، وتقنيات الاتصال والمعلومات التي تساعد في المراقبة الصحية المستمرة والاكتشاف الفوري للمشكلات الطبية أو الحالات الطارئة.

يهدف هذا المشروع إلى تطوير سوار إلكتروني ذكي مزود بحساسات لقياس المؤشرات الحيوية، بالإضافة إلى وحدة GPS وبطاقة SIM لربط بيانات المريض بموقع إلكتروني مخصص. يتيح السوار للمشرفين مراقبة حالة المريض في الوقت الفعلي، حيث يوفر معلومات أساسية مثل الموقع، والحالة الصحية، والتنبيهات في حالات الطوارئ.

يقوم السوار بإرسال التنبيهات في الحالات الحرجة، كما يتيح الوصول السريع إلى بيانات المريض من خلال مسح رمز QR. يهدف هذا المشروع إلى تعزيز مراقبة صحة المرضى، وتسريع الاستجابة للطوارئ، وتحسين خدمات الرعاية الصحية عن بُعد.

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Chapter one

Introduction

Chapter One: Introduction

1.1 Introduction

Tracking Bracelet is A Life-Saving Technology for People with Special Needs For families caring for individuals with special needs, the fear of losing them or them encountering emergencies is a constant source of anxiety. Despite ongoing efforts to monitor and protect them, unexpected situations can arise, leaving families feeling helpless and worried.

This is where the electronic tracking bracelet comes in, combining GPS technology to track real-time location with a heart rate sensor to continuously monitor the user's health. The bracelet is linked to a dedicated website for the caregiver, allowing them to track the patient's movements and health status through displayed data of location and heart rate. This information is updated automatically on the website, which also contains essential details like the patient's name, health condition, and the caregiver's phone number. Additionally, the bracelet includes a QR Code that, when scanned, directs to an identification webpage displaying the patient's critical information, helping expedite actions in emergency situations.

The use of the tracking bracelet is not limited to children with special needs but also extends to adults suffering from medical conditions such as Alzheimer's or dementia. This technology provides peace of mind for families and helps save lives.

1.2 Problem Statement

The Disability Support Bracelet is a wearable device equipped with NFC technology, enabling caregivers or authorized personnel to access the user's critical medical and emergency information through a secure website. It is designed without a screen for simplicity, with all data securely stored online for easy retrieval. By providing quick access to vital information, it enhances safety, supports caregiver coordination, and promotes the independence of persons with disabilities.

Moreover, the inclusion of a QR Code on the bracelet allow instant access to the patient's identity card, which contains essential details such as their name, medical condition, and emergency contact number.

1.3 Research Objectives

Three main objectives direct this study as follows

- To explore the state-of-the-art literature on simulating a model for communication between sensors and the Arduino controller to collect vital data for patients.
- To develop an electronic bracelet equipped with GPS and a heart rate sensor to monitor the real-time location and health status of individuals with special needs.
- To implement a dedicated website and QR Code system that allows caregivers to access patient information, receive emergency alerts, and update health data in real time.

1.4 Project Methodology

The methodology followed in this project has two parts

Firstly, Hardware Components Includes the bracelet's components such as the QR Code, heart rate sensor, GPS module, Arduino, and communication modules.

Secondly Software Development Involves the web application, Arduino code, and the backend system that processes the patient's health and location data.

1.5 Layout project the rest of study organized to five chapters

- Chapter one: introduction.
- Chapter two: Theoretical Background and Literature Review.
- Chapter three: System Design and Analysis

- Chapter four: Methodology.
- Chapter Five: Conclusion and Recommendations.

Chapter Two

Theoretical Background and Literature Review

Chapter Two: Literature Review and Background

2.1.1 Introduction

In this chapter, we discussed all the information related to the project and present previous studies.

2.1.2 People with Special Needs

The term "people with special needs" refers to individuals who face physical, mental, psychological, or sensory challenges that require specific care or additional support to adapt and integrate effectively into society [1].

2.1.3 Physical Disabilities

This category includes individuals who experience partial or complete loss of mobility or physical functions due to injuries or chronic health conditions. These individuals may rely on wheelchairs, prosthetics, or other assistive devices to maintain their independence [2].

2.1.4 Sensory Impairments

People with sensory impairments suffer from partial or complete loss of hearing or vision. Support for this group includes tools such as hearing aids, optical glasses, and Braille systems to ensure accessibility and independence [3].

2.1.5 Intellectual Disabilities

This category comprises individuals with conditions such as autism, Down syndrome, and learning disabilities. These individuals face challenges in learning and social interaction, necessitating adaptive teaching methods and tailored support environments to meet their needs [4].

2.1.6 Psychological Disorders

This group includes individuals with mental health issues such as depression, anxiety, and post-traumatic stress disorder (PTSD). These conditions can significantly impact mental well-being and social interactions, requiring therapeutic interventions and consistent psychological support [5].

2.1.7 Alzheimer's disease

is a neurodegenerative condition leading to a decline in memory, thinking, and behavior. It significantly affects the ability to perform daily activities and has profound social and emotional impacts [6].

2.1.8 Tracking

Tracking is the process of monitoring or following the movement or behavior of a person or object over a defined period to gather precise information about its location, behavior, or condition. It is widely utilized in fields such as healthcare, security, business, and social sciences. The methods and tools for tracking vary depending on its purpose and may include modern technologies like Global Positioning Systems (GPS), wearable devices, and digital software [7].

2.1.9 Types of Tracking

2.1.9.1 Geographic Tracking

Geographic tracking refers to the use of technologies such as GPS to monitor the geographical locations of individuals or objects. This type of tracking is used in applications like navigation systems, fleet management, and tracking individual movements during emergencies. It is a vital tool in logistics and delivery industries, enhancing efficiency and reducing delivery time [8].

2.1.9.2 Health Tracking

Health tracking involves the use of devices like smartwatches and fitness trackers to monitor physiological and health-related data. It enables the collection of information such as physical activity, heart rate, and sleep patterns, which can be used to improve quality of life and provide timely medical interventions. This type of tracking is increasingly used in medical research and personalized healthcare to monitor and manage chronic conditions [9].

2.1.9.3 Digital Tracking

Digital tracking monitors individuals' online activities, such as visited websites, clicks, and submitted data. It is widely applied in digital

marketing to analyze consumer behavior and customize advertisements based on individual interests. Digital tracking allows the collection and real-time analysis of large volumes of data, offering precise insights into user preferences and behavior patterns [10].

2.1.9.4 Security Tracking

Security tracking involves monitoring individuals or assets for safety purposes. It may include tracking the movements of important individuals, monitoring sensitive shipments, or gathering evidence in criminal investigations. Studies suggest that security tracking enhances the ability of security agencies to respond swiftly to emergencies and improve risk management [11].

2.1.10 Practical Applications of Tracking

Tracking serves as a critical tool in various fields. In healthcare, tracking monitors patients and improves chronic disease management, such as for heart disease and diabetes [12]. In business, tracking optimizes resource management and reduces costs by monitoring employee performance and asset movements [13]. In security, tracking helps protect individuals and assets through effective monitoring of suspicious activities and crime prevention [11].

2.1.11 the Global Positioning System (GPS)

The Global Positioning System (GPS) is a satellite-based navigation system that allows users to determine their geographical locations with high accuracy anywhere on Earth. Developed by the United States Department of Defense in the 1970s for military purposes, it became available for civilian use in the 1980s. GPS relies on a network of satellites that transmit precise signals to provide data on location and time. These signals are received by GPS receivers on the ground, which analyze the signals to determine the geographical position of the user [14].

2.1.12 A heart rate sensor

is a device used to measure and record an individual's heart rate. It relies on sensing technologies that detect changes in blood flow resulting from heartbeats. One of the most common types of these sensors is the optical

sensor, which uses infrared (IR) technology to measure blood flow in blood vessels. [15]

2.1.13 Components of the GPS System

The GPS system consists of three main components

2.1.13.1 The Space Segment

This includes the satellites orbiting Earth in predetermined trajectories. The GPS system comprises 24 to 32 active satellites continuously transmitting signals that cover the entire Earth's surface.

2.1.13.2 satellites The Ground Segment

This consists of ground control stations used to monitor the and adjust their orbits. These stations update the navigation data and signals transmitted by the satellites to ensure their accuracy.

2.1.13.3 The User Segment

This includes GPS receivers found in vehicles, smartphones, and other devices. These receivers analyze the signals transmitted by the satellites to determine geographical location, speed, and time with high accuracy [8].

2.1.14 Types of GPS Systems

2.1.14.1 Conventional GPS (Basic GPS)

This is the basic type of GPS primarily used in civilian applications such as navigation in cars and smartphones. This system relies on signals received from three to four satellites to determine location with an accuracy of a few meters. It is used in applications that do not require high precision, such as guiding drivers and urban navigation [16].

2.1.14.2 Differential GPS (DGPS)

The Differential GPS system provides higher accuracy compared to the conventional GPS, reducing location errors to less than one meter. DGPS relies on ground-based reference stations that send correction signals to GPS receivers, enhancing location accuracy. It is widely used in marine navigation, precise mapping, and geological research [17].

2.1.14.3 Wide Area Augmentation System (WAAS)

The WAAS is a type of GPS designed specifically for civil aviation in the United States. WAAS provides improvements in location accuracy by using ground stations that monitor satellite signals and send corrections to geostationary satellites. This system offers an accuracy of less than three meters, making it suitable for aviation applications [18].

2.1.14.4 Real-Time Kinematic GPS (RTK GPS)

RTK GPS is one of the most accurate positioning tools, capable of achieving accuracy within centimeters. RTK uses carrier wave measurements and relies on fixed reference stations that send corrective signals to mobile receivers. RTK is used in civil engineering, precision agriculture, and projects requiring very high accuracy in location determination [19].

2.1.15 Practical Applications of GPS

GPS is used in a wide range of daily and professional applications. In the civil sector, GPS is an essential tool for navigation in cars, airplanes, and ships, facilitating mobility and improving efficiency. It is also employed in scientific research for topographic mapping, earthquake studies, and monitoring land changes. In the military field, GPS is used to guide missiles and vehicles and accurately locate targets. Studies indicate that the accuracy, ease of use, and availability of the GPS system around the clock make it one of the most important technologies of the twenty-first century [20].

2.1.16 Barcode

A barcode is an encoding system used to convert information into a machine-readable format using scanners. It consists of a sequence of lines or bars of varying sizes that represent digital or textual data in a manner that can be accurately and quickly interpreted by devices. Barcodes are

primarily used in commerce, inventory management, and supply chains to facilitate product identification and tracking, thereby enhancing operational efficiency [21].

2.1.17 A Subscriber Identity Module Card (SIM card)

is a small removable smart card used in mobile devices to store information related to the subscriber's identity, such as their phone number and network details. It allows the device to connect to the cellular network by providing the necessary authentication and encryption keys for secure communication. [22]

2.1.18 Biosensors

Biosensors are analytical devices that utilize biological elements such as enzymes or antibodies to interact with a specific substance, converting these biological interactions into measurable electrical signals through a transducer. These devices are widely applied in medical fields for early disease diagnosis, as well as in environmental monitoring and food safety assurance [23].

2.1.19 Data Security and Privacy

Data security and privacy refer to a set of measures and technologies designed to protect data from unauthorized access, modification, disclosure, or destruction. Data security focuses on technical safeguards such as encryption, passwords, and firewalls to ensure data protection. Privacy, on the other hand, emphasizes safeguarding individuals' personal information from misuse or unauthorized access, ensuring that such data is utilized in accordance with established policies and the consent provided [24].

2.1.20 Arduino

Arduino is an open-source electronics platform used for building interactive electronic projects. It consists of a microcontroller board and software (Integrated Development Environment – IDE) used to write and upload code to the board. Arduino allows users to easily create electronic projects through programming in the simple C/C++ language and connecting various electronic components. It is widely used in research and development, hobbies, and education [25, 26].

2.1.21 Types of Arduino Boards

There are several types of Arduino boards, each with distinct features and specifications suited for different needs. Below are some of the main types

2.1.21.1 Arduino Uno

This is the most popular and beginner-friendly board. It uses the ATmega328P microcontroller and has 14 digital I/O pins (6 of which can be used as PWM), 6 analog input pins, a USB connection for programming, and a power connector [25].

2.1.21.2 Arduino Mega 2560

This board is equipped with more pins, making it ideal for larger, more complex projects. It contains 54 digital I/O pins (15 of which can be used as PWM), 16 analog input pins, and 4 UART ports for serial communication. It uses the ATmega2560 microcontroller [25].

2.1.21.3 Arduino Due

The first Arduino board to use the ARM Cortex-M3 32-bit microcontroller, offering higher processing power and speed compared to other boards. It includes 54 digital I/O pins and 12 analog input pins [25].

2.1.21.4 Arduino Nano

A small-sized board suitable for projects requiring compact space. It uses the ATmega328P or ATmega168 microcontroller and has 14 digital I/O pins and 8 analog input pins [27].

2.1.21.5 Arduino Leonardo

This board is capable of emulating USB devices such as keyboards or mice, thanks to the ATmega32u4 microcontroller. It features 20 digital I/O pins and 12 analog input pins [28].

2.1.22 Databases

Databases are organized collections of data stored in a manner that facilitates efficient management, retrieval, and updating. They are used to structure information for easier access and analysis and are essential in various applications such as banking systems, inventory management, and websites. Databases are managed using Database Management Systems (DBMS), which provide tools for creating, querying, and updating data [29].

2.1.23 A database schema

is the organizational structure that defines how data is stored in a database. This structure includes tables, columns, relationships between tables, and the keys that link them, along with the constraints applied to the data to ensure consistency and reliability. The schema defines how data is organized within the system and how information is interrelated, enabling easy retrieval and analysis of data. [30].

2.1.24 Principles of Management

A supervisor is defined as an individual who is responsible for organizing and guiding work teams to achieve defined goals. Their role includes providing support and training, managing performance, and developing strategies to improve effectiveness and efficiency [31].

2.1.25 Website

A website is a collection of interlinked web pages stored on a single web server, accessible online through a web browser. Each website is identified by a unique URL, often registered with its own domain name. These pages consist of text, images, videos, and other multimedia, which are connected through hyperlinks to provide a navigable experience for the user [32].

Websites serve various purposes, such as informational, commercial, educational, social, or entertainment-related. The basic structure of a website includes a homepage that links to different subpages. These pages can be created using programming languages like HTML, CSS, and JavaScript, and supported by technologies such as databases to manage content [33].

Designing and developing a website is a crucial part of web development, requiring knowledge of diverse web technologies to ensure a good user experience and secure data [34].

2.1.26 Tracking Bracelet

A tracking bracelet, also known as a smart band, is an electronic wearable device designed to monitor and record the user's physiological and behavioral data. These devices are typically equipped with various sensors, such as accelerometers, heart rate monitors, and motion sensors, enabling the tracking of daily activities, including step count, heart rate, distance traveled, and calories burned. Additionally, tracking bracelets can monitor sleep patterns and analyze overall sleep quality [35].

The data collected by these devices in real-time is often transmitted to smartphone applications or stored on cloud platforms for analysis. This analysis provides detailed insights into an individual's health status, aiding in the achievement of personal health and fitness goals [36].

Tracking bracelets are employed in various medical and athletic applications, including chronic disease prevention, performance enhancement, and quality-of-life improvement. Systematic reviews demonstrate the high accuracy and reliability of these devices in measuring physical activities, supporting their use in scientific research and health monitoring [37].

2.1.27 Innovative Solutions Provided by Smart Wristbands

Smart wristbands offer innovative solutions to numerous health challenges faced by individuals, particularly those with special needs or chronic health conditions. Below are some of the key solutions provided by these devices

2.1.27.1 Monitoring Vital Signs

Smart wristbands provide continuous monitoring of vital signs such as heart rate, blood pressure, and blood oxygen levels. This facilitates early detection of abnormal changes that may indicate unstable health conditions [38]

2.1.27.2 Managing Physical Activity

These wristbands track physical activities such as step count, calories burned, and sleep patterns. This enhances weight management and fitness improvement, potentially reducing the risk of chronic diseases [39].

2.1.27.3 Emergency Alerts

Some smart wristbands feature emergency alert systems, sending notifications in critical situations such as falls or abnormal heart rate spikes, enabling quick medical assistance [40].

2.1.27.4 Integration with Health Apps

Smart wristbands can integrate with smartphone health applications, providing detailed health reports that assist doctors in monitoring conditions and adjusting treatment plans if needed [41].

2.1.27.5 Monitoring Activity and Therapy

These wristbands aid in tracking therapeutic activities, such as prescribed exercises for individuals with chronic conditions like diabetes or heart diseases, enhancing treatment efficacy [42].

2.1.27.6 Personalized Medication Tracking

Some wristbands can issue reminders for medication intake and record adherence data, improving treatment outcomes and management of chronic conditions [43].

2.1.27.7 Stress Management

Certain smart wristbands can measure stress levels through indicators such as heart rate variability and provide strategies for stress reduction, including deep breathing exercises and relaxation techniques [44].

2.2 Literature Review

2.2.1 Overview of Related Works

This chapter presents a review of previous studies and projects that align with the goals of the "Persons with Disabilities Bracelet (PWDB)," which integrates GPS and vital signs monitoring for people with disabilities. Below is an analysis of related works to contextualize and differentiate the project.

J. Schlingensiepen et al. [45] discusses Empowering People with Disabilities Using Urban Public Transport Published in Procedia Manufacturing (2015) This study developed a system for assisting individuals with disabilities in navigating public transport systems. It employed personalized navigation tools using GPS, RFID, and Bluetooth technologies, The system provided real-time guidance, such as identifying the correct bus stop or vehicle and ensuring the traveler reached their destination safely, Relevance The bracelet in the "mobile" project utilized RFID and Bluetooth for localization, similar to the PWDB's integration of GPS for locating users in emergencies.

Tanweer Alam et al. [46] developed Designing a People Tracking System for Smart Cities Published in International Journal of System Assurance Engineering and Management (2021) This paper introduced a GPS-based tracking application designed for crowded environments, enabling guardians to monitor vulnerable individuals such as children and the elderly. The system incorporated geofencing and real-time alerts for location tracking. Relevance Highlights the use of geofencing in ensuring users' safety, a concept adaptable to PWDB for defining safe zones and alerting supervisors.

Marina Bolsunovskaya et al. [47] discusses Innovative Project Model Development for Wearable Devices Published in Springer Lecture Notes (2022) This research examined wearable devices for health monitoring, focusing on data transmission and encryption to ensure secure

communication between devices and mobile applications. Functionalities included monitoring heart rate, temperature, and activity levels, with real-time data sent to a synchronized application. Relevance Aligns with PWDB's features of real-time vital signs monitoring and secure data communication.

Raghavan et al. [48] Child Tracking System Using Android Phones Published in International Journal of System Assurance Engineering and Management (2020) Proposed an Android-based GPS and SMS system for tracking lost children. The app displayed real-time locations and provided location-sharing capabilities. Relevance Demonstrates the practical implementation of GPS and mobile app integration, a critical component of the PWDB project.

2.2.2 Technologies Leveraged

Common technologies include GPS for location tracking, RFID for short-range identification, Bluetooth for connectivity, and geofencing for safety zones.

2.2.3 Data Security

Emphasis on secure data transmission and privacy, crucial for wearable health-monitoring devices.

2.2.4 Personalized Assistance

Projects like "mobile" underline the importance of customization to meet individual needs, such as specific health conditions or mobility constraints.

2.2.5 Practical Applications

Tracking systems for children, the elderly, and people with disabilities demonstrate the feasibility of similar solutions, validating the approach of PWDB.

Chapter Three

System Design and Analysis

Chapter Three: System Design and Analysis

3.1 Introduction

This chapter explains functional and non-functional requirements and the analysis of the system using the Unified Modelling Language (UML) models.

3.2 Functional Requirement

The following requirements describe how the system should function, including the features it must provide.

3.2.1 Barcode Data Reading

The system must be capable of reading the barcode on the patient's wristband to extract basic patient data, such as name, age, and medical ID.

The system must extract the supervisor's ID from the barcode, enabling the caregiver to contact them.

3.2.2 Sensor Data Reading

The system must read real-time heart rate data from the sensors.

The system must read real-time geographical location data from the GPS sensor.

3.2.3 Data Display on Supervisor's Webpage

The patient's heart rate and geographical location must be displayed in real-time on the supervisor's webpage.

3.2.4 Heart Rate Data Analysis

The system must analyze heart rate data to detect potential issues, such as arrhythmia or tachycardia.

3.2.5 Hazard Alert to Supervisor

In case of a heart rate anomaly or emergency, the system must send a hazard alert including the patient's geographical location to the supervisor using appropriate communication means (e.g., SMS or in-app notification).

3.2.6 Caregiver-Supervisor Communication

The caregiver must be able to communicate with the supervisor using the supervisor's ID extracted from the barcode.

3.3 Non-Functional Requirements

These requirements define how the system should perform, such as performance, reliability, and usability.

3.3.1 System Performance

The system must be capable of reading sensor data (heart rate and GPS) in real-time without delay.

Patient data (e.g., heart rate and location) must be displayed on the supervisor's webpage within 2-3 seconds of receipt.

3.3.2 Reliability

The system must be reliable in monitoring the patient's condition at all times.

The system must be able to send hazard alerts in case of a heart rate anomaly promptly, so that notifications reach the supervisor within 30 seconds of the anomaly.

3.3.3 Security

Patient data must be protected using encryption techniques to ensure personal data confidentiality.

Communication between system devices (e.g., reader device, wristband, webpage) must be secured to ensure data privacy.

3.3.4 Availability

The system must be available for operation 24/7 without interruption or sudden shutdown, ensuring continuous patient monitoring.

The webpage displaying patient data must be stable and operate efficiently.

3.3.5 Usability

The supervisor's webpage interface must be simple and user-friendly, allowing the supervisor to quickly access displayed data.

The barcode reader device interface must be simple and user-friendly for caregivers.

3.3.6 Scalability

The system must be scalable to support a larger number of users in the future, such as adding more sensors or monitoring systems.

3.3.7 Emergency Response

The system must be able to handle emergency situations quickly and effectively, including sending immediate notifications to the supervisor.

3.4 system modelling

Modelling system by using different UML models

3.4.1 Flowchart diagram model

Therefore, to represent the flow of processes or data flow within a specific system. The sequential steps that occur in the system are clearly shown, which helps in understanding how the system works.

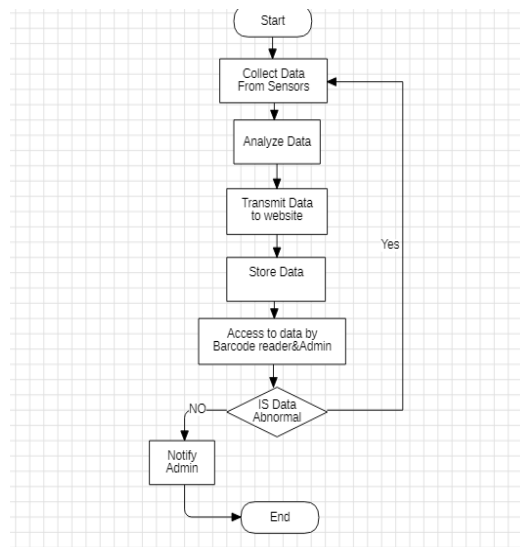


Figure 3.1: bracelet system flowchart

3.4.2 Use-case diagram model

It is used to represent how users (Actors) interact with the system to achieve certain goals. This diagram shows the processes or functions of the system that are understandable to end users.

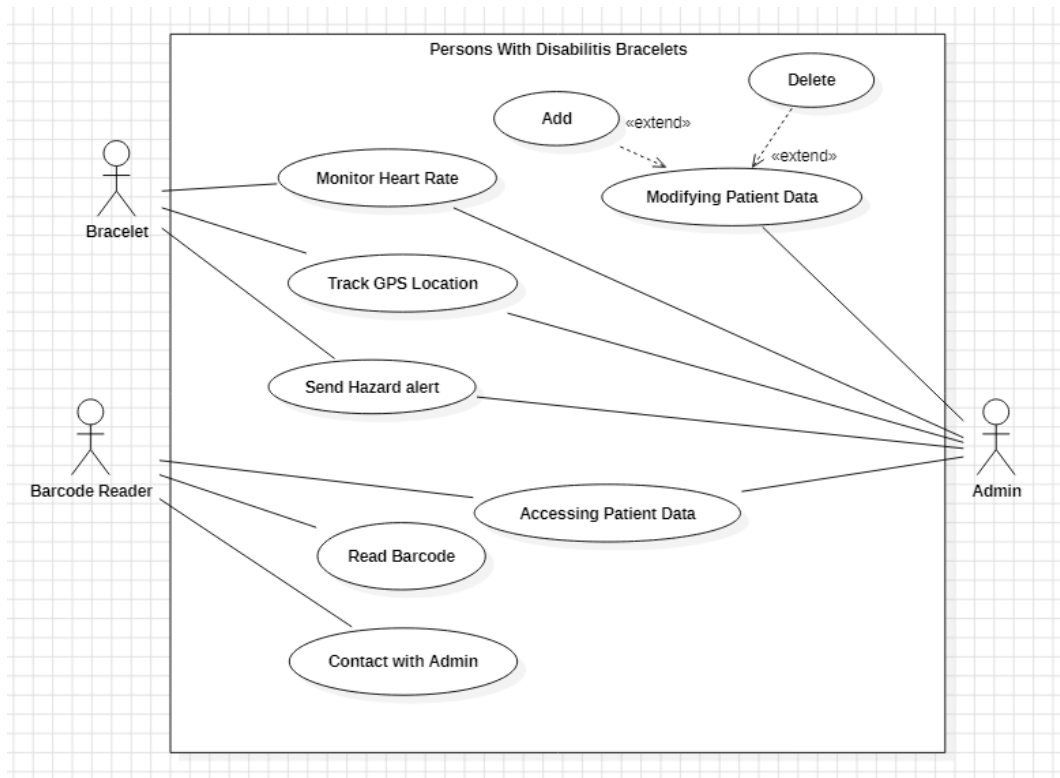


Figure 3.2: bracelet system use case

3.4.3 Dataflow diagram model

This diagram aims to illustrate how data moves within the system, and how it is processed by different components (such as processes, stores, and external sources). A data flow diagram is used in systems analysis to depict how the system works overall and what data is circulated between components.

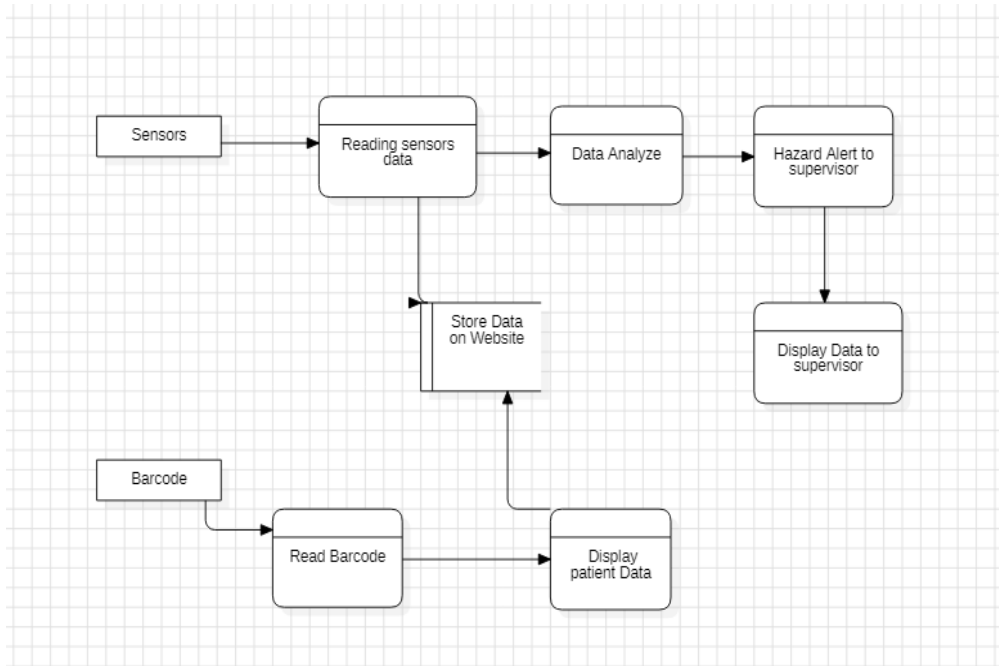


Figure 3.3: bracelet system dataflow

3.5 Database Patient information

Column name	Data Type	Constraint	Description
Patient name	VARCHAR (100)	NOT NULL	Full name of the patient
Patient	INT	PRIMARY KEY	Unique identifier for each patient
Patient address	VARCHAR (255)	NOT NULL	Address of patient
Supervisor phone	VARCHAR (15)	NOT NULL	Phone number of the supervisor

Table 3.1: Patient

3.6 Challenges faced by the project

- Difficulty in obtaining hardware components

Due to the current conditions in the country, it was not possible to obtain the physical components needed to implement the project, which led us to rely on the simulation.

- Limitations and lack of integration of tools

The simulation that was used was limited in simulating all the required components and scenarios, and we also faced difficulty in linking the simulation with Website to display the data live, due to the lack of support for the tools used for this.

- Technical challenges in connecting the simulation to external applications

The Proteus program did not support physical connectivity with external applications such as Website, which led to the inability to display live data readings on the admin page.

Chapter Four

Result and discussion

Chapter Four: Result and discussion

4.1 Introduction

This chapter explains the implementation of the Persons with Disabilities Bracelet (PWDB) system through three steps

- The hardware components, including the Arduino, heart rate sensor, GPS module, SIM900 GSM module, LED indicator, and a variable resistor, are connected and simulated in Proteus 8.
- The Arduino code is written and uploaded to process heart rate data, detect abnormalities, and send emergency alerts via the SIM900 module to a random caregiver.
- A web application is developed to display patient data, including name, location, heart rate, and caregiver contact information. A QR code is also generated and printed on the bracelet, linking to the patient's identification page for quick access in emergencies.

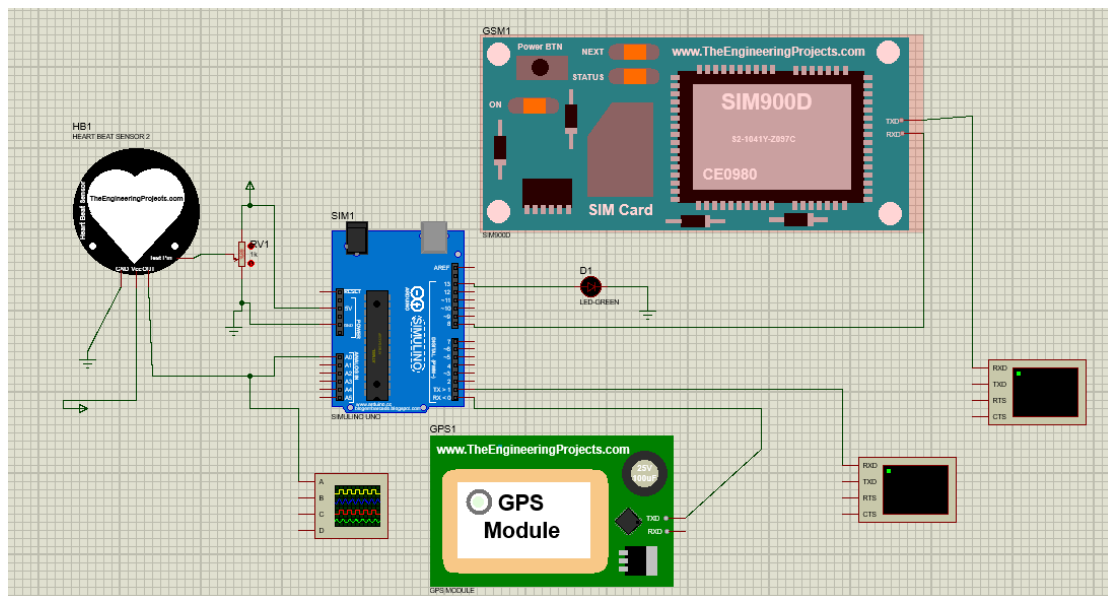


Figure 4.1 Hardware Circuit Diagram

4.2 Persons with Disabilities Bracelet (PWDB)

4.2.1 Arduino Uno – The Brain of the System

The Arduino Uno is the central microcontroller responsible for processing and managing all components in the bracelet. It collects data from the heart rate sensor, analyzes it, and determines if an emergency alert should be sent. It also communicates with the GPS module to get the user's location and controls the SIM900 module to send SMS alerts when necessary.

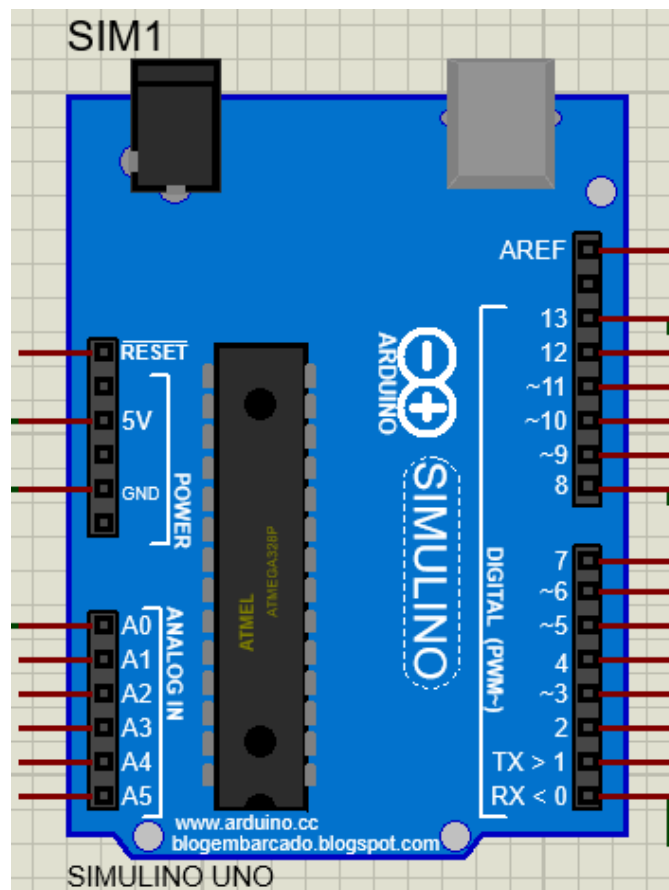


Figure 4.2: Arduino Uno

4.2.2 The Heart Rate Sensor

Is responsible for continuously measuring the pulse and sending the data to the Arduino Uno for processing.

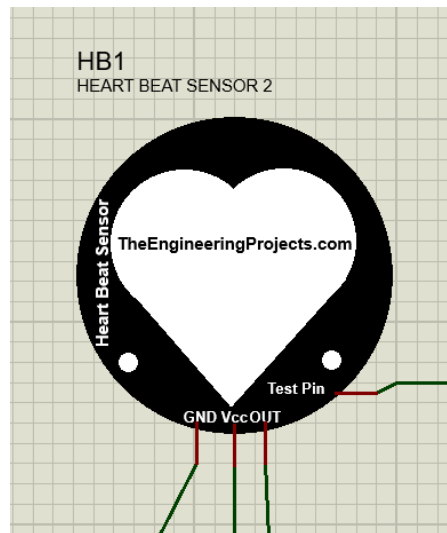


Figure 4.3: Heart Rate Sensor

4.2.3 GPS Module

The GPS Module tracks the patient location and sends the coordinates to the Arduino Uno, when an abnormal heart rate is detected.

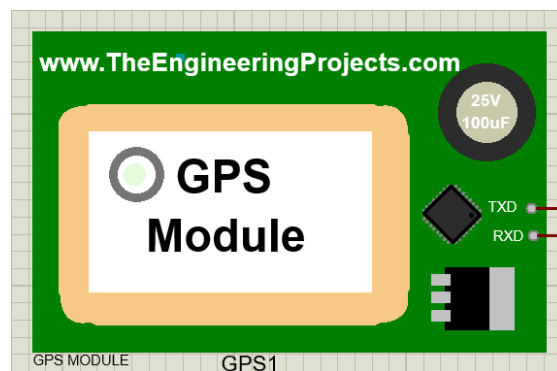


Figure 4.4: GPS Module

4.2.4 SIM900 GSM Module

The SIM900 GSM Module is responsible for sending emergency SMS alerts to caregivers when an abnormal heart rate is detected. It communicates with the Arduino Uno, which processes the heart rate and GPS data, and then instructs the SIM900 to send an SMS containing the user's heart rate and real-time location. This allows caregivers to respond quickly in emergencies.

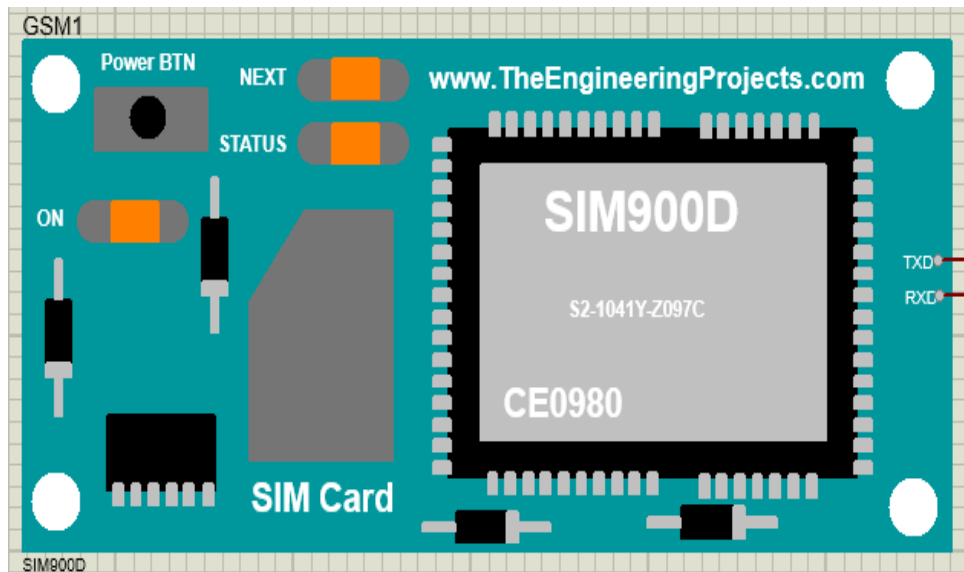


Figure 4.5: GSM SIM900 Module

4.2.5 Virtual Terminal Screen

The Virtual Terminal Screen serves as a display for emergency alerts generated by the system. It shows notifications when an abnormal heart rate is detected, including the patient's current BPM and their GPS coordinates. And the caregiver phone number that the alerts was sanded to.

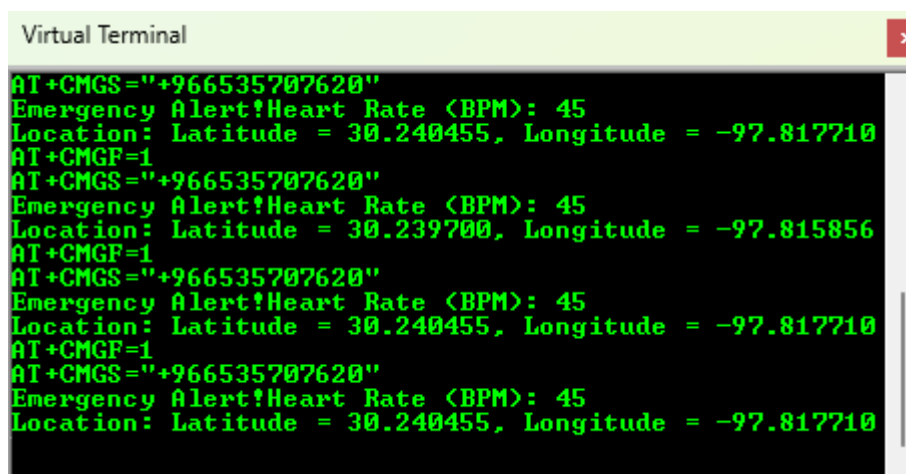


Figure 4.6: Virtual Terminal Screen

4.2.6 Variable Resistor

The Variable Resistor here is used to simulate heart rate fluctuations by adjusting the input voltage to the Heart Rate Sensor. The variable resistor mimics heart rate variations by altering the sensor's output, allowing the system to detect changes in pulse rate. It acts as a manual control for generating different heartbeat scenarios, helping validate the system's response to critical conditions.

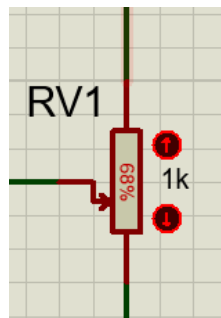


Figure 4.7: Variable Resistor

4.2.7 Green LED

The Green LED serves as a visual indicator that lights up whenever a heartbeat is detected, regardless of whether it falls within the normal range or not.

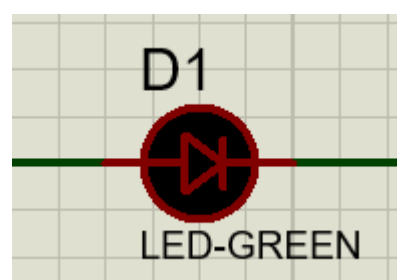


Figure 4.8: Green LED

4.2.8 The Oscilloscope

The Oscilloscope is connected to the Heart Rate Sensor to visually display the detected heartbeats as waveforms.

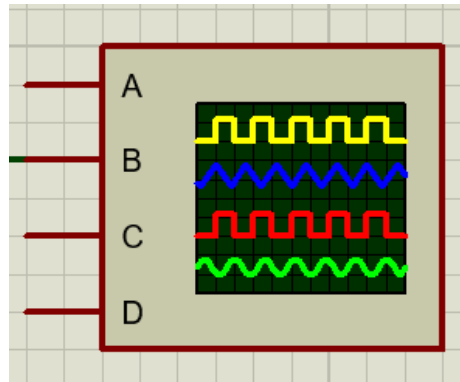


Figure 4.9: The Oscilloscope

4.2.9 The web application

The web application shown is designed for caregivers to remotely monitor patients wearing the Persons with Disabilities Bracelet (PWDB). It displays key patient data, including name, address, phone number, heart rate, and current location. There is also an "Update Data" button, allowing caregivers to modify patient information. Below, a brief description explains the purpose of the electronic bracelet.

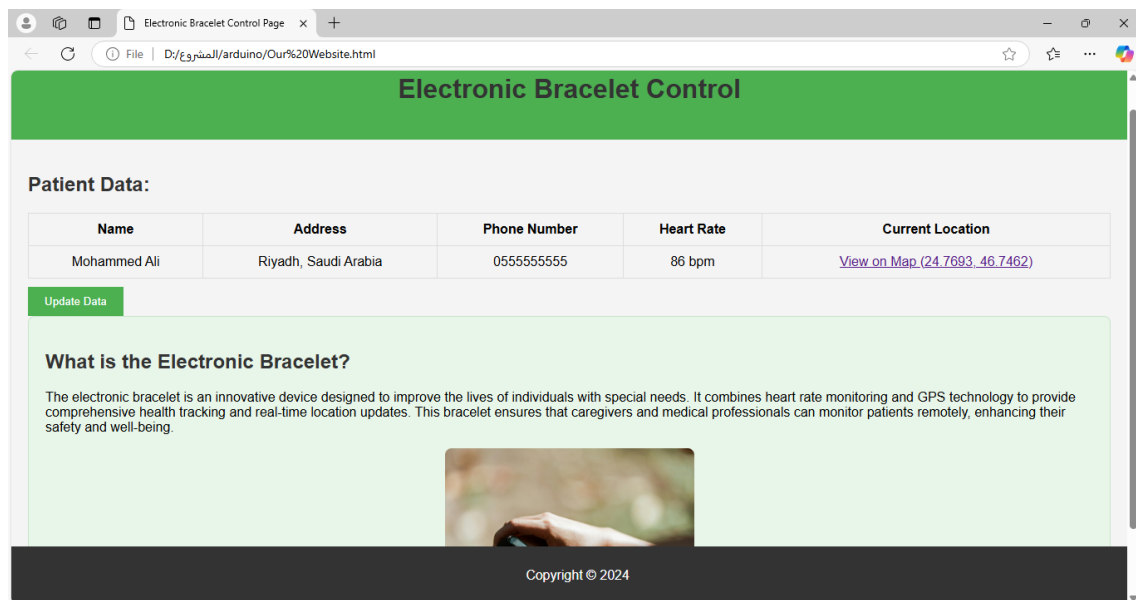


Figure 4.10: The web application

4.2.10 QR Code

A QR code on each patient's bracelet, which leads to a website displaying their identity card. This feature helps in quickly identifying and assisting patients in case of emergencies.



Figure 4.11: QR Code

4.2.11 Identity card in a website

Is the website linked to the QR code on the bracelet serves as an identification platform for the patient. It displaying an identity card or Patient information includes the patient's name, their medical condition, the caregiver's phone number, and their place of residence.

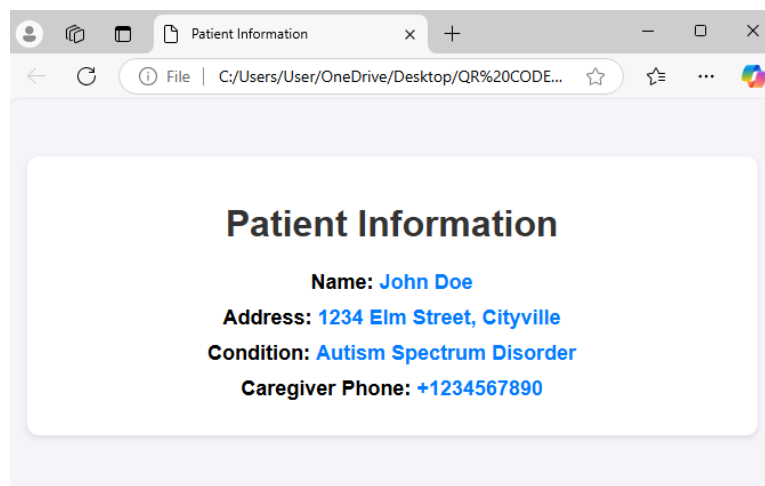


Figure 4.12: Identity card

Chapter Five Conclusion and Recommendations

Chapter Five: Conclusion and Recommendations

5.1 Introduction

This chapter discusses the key findings of the Persons With Disabilities Bracelet (PWDB) project. It highlights the system's effectiveness, discusses limitations, and suggests future improvements to enhance functionality and usability.

5.2 Conclusion

In this project, we designed a tracking bracelet specifically for individuals with special needs. The bracelet includes a heart rate sensor to monitor the user's health condition, as well as a GPS sensor to share the user's location with the administrator (Admin) via a SIM card. In case of an abnormal increase or decrease in heart rate, the system sends an emergency alert to the administrator's phone number. Additionally, the bracelet features a printed QR code containing the user's information and the administrator's phone number. A dedicated website is also available for the administrator to monitor the user's health status and track their location.

5.3 Study Contributions

This study contributes to the healthcare sector by providing a cost-effective and efficient solution for monitoring individuals with special needs, preventing cases of loss, and facilitating health status tracking.

5.4 Recommendations

It is recommended to add more sensors to enhance the system's usability, scalability, and integration with other healthcare systems. Additionally, converting the website into a mobile application could improve efficiency and simplify the administrator's tasks.

5.5 Final Thoughts

This project highlights the potential of tracking technologies and biometric sensors in improving the quality of life for individuals with special needs, making healthcare monitoring and supervision more accessible and effective.

5.6 References

- [1] World Health Organization. (2011). World Report on Disability. World Health Organization. <https://www.who.int/publications/i/item/9789241564182>
- [2] Harrison, T. C. (2006). Living with Severe Physical Impairment: The Experience of Young Adults with Complex Disabilities and Their Families. *Disability & Society*, 21(4), 361–374. <https://doi.org/10.1080/09687590600679911>
- [3] National Institute on Deafness and Other Communication Disorders. (2017). Hearing Loss and Older Adults. Retrieved from NIDCD website. <https://www.nidcd.nih.gov/health/hearing-loss-older-adults>
- [4] American Psychiatric Association. (2013). Diagnostic and Statistical Manual of Mental Disorders (5th ed.). <https://doi.org/10.1176/appi.books.9780890425596>
- [5] Kessler, R. C., Chiu, W. T., Demler, O., & Walters, E. E. (2005). Prevalence, Severity, and Comorbidity of 12-Month DSM-IV Disorders in the National Comorbidity Survey Replication. *Archives of General Psychiatry*, 62(6), 617–627. <https://doi.org/10.1001/archpsyc.62.6.617>
- [6] World Health Organization. (2017). Dementia. Retrieved from. <https://www.who.int/news-room/fact-sheets/detail/dementia>
- [7] Dodge, S., Weibel, R., & Lautenschütz, A. K. (2008). Towards a taxonomy of movement patterns. *Information Visualization*, 7(3-4), 240-252. <https://doi.org/10.1057/palgrave.ivs.9500182>

- [8] El-Rabbany, A. (2002). Introduction to GPS: The Global Positioning System. Artech House.
<https://www.artechhouse.com/Introduction-to-GPS-The-Global-Positioning-System-Second-Edition-P9781580531832.aspx>
- [9] Piwek, L., Ellis, D. A., Andrews, S., & Joinson, A. N. (2016). The rise of consumer health wearables: Promises and barriers. *PLOS Medicine*, 13(2), e1001953.
<https://doi.org/10.1371/journal.pmed.1001953>
- [10] Zuboff, S. (2019). The age of surveillance capitalism: The fight for a human future at the new frontier of power. PublicAffairs.
<https://www.publicaffairsbooks.com/titles/shoshana-zuboff/the-age-of-surveillance-capitalism/9781610395694/>
- [11] Gill, M., Bryan, J., & Allen, J. (2007). Public perceptions of CCTV in residential areas: “It is not as good as we thought it would be”. *International Criminal Justice Review*, 17(4), 304-324.
<https://doi.org/10.1177/1057567707311596>
- [12] Swan, M. (2012). Health 2050: The realization of personalized medicine through crowdsourcing, the quantified self, and the participatory biocitizen. *Journal of Personalized Medicine*, 2(3), 93-118.
<https://doi.org/10.3390/jpm2030093>
- [13] Wamba, S. F., Anand, A., & Carter, L. (2013). A literature review of RFID-enabled healthcare applications and issues. *International Journal of Information Management*, 33(5), 875-891.
<https://doi.org/10.1016/j.ijinfomgt.2013.07.005>
- [14] Kaplan, E. D., & Hegarty, C. J. (2006). Understanding GPS: Principles and Applications. Artech House.
<https://www.artechhouse.com/Understanding-GPS-Principles-and-Applications-Second-Edition-P9781580538947.aspx>
- [15] Wright, S. P., Hall Brown, T. S., Collier, S. R., & Sandberg, K. (2017). How consumer physical activity monitors could transform human physiology research. *American Journal of Physiology-Regulatory, Integrative and Comparative Physiology*, 312(3),

R358-R367.

<https://doi.org/10.1152/ajpregu.00349.2016>

- [16] Hofmann-Wellenhof, B., Lichtenegger, H., & Collins, J. (2007). Global Positioning System: Theory and Practice. Springer.
<https://www.springer.com/gp/book/9783211433810>
- [17] Langley, R. B. (1991). The GPS Observables. GPS World, 2(4), 52-59.
<https://www.gpsworld.com/the-gps-observables/>
- [18] Enge, P., & Walter, T. (1996). Wide Area Augmentation System. In Global Positioning System: Theory and Applications (pp. 193-235). AIAA.
<https://searchworks.stanford.edu/view/3104510>
- [19] El-Mowafy, A. (2000). Real-Time Kinematic GPS (RTK GPS). University of Calgary, Department of Geomatics Engineering.
<https://schulich.ucalgary.ca/geomatics/research/research-areas/graduate-student-research>
- [20] Misra, P., & Enge, P. (2006). Global Positioning System: Signals, Measurements, and Performance. Ganga-Jamuna Press.
<https://www.amazon.com/Global-Positioning-System-Applications-Astronautics/dp/156347249X>
- [21] Biedenkapp, D. (2010). Barcodes: Principles and Applications. Springer.
https://link.springer.com/chapter/10.1007/978-3-642-59742-8_54
- [22] Pahlavan, K., & Li, X. (2011). Wireless Communications: Principles and Practice (2nd ed.). Prentice Hall.
https://link.springer.com/chapter/10.1007/978-3-642-59742-8_54
- [23] Turner, A. P. F. (2000). Biosensors: Sense and sensibility. Chemical Society Reviews, 29(4), 169-176.
https://link.springer.com/chapter/10.1007/978-3-642-59742-8_54
- [24] Lin, Xiaodong, and Rongsheng Luo. Security and Privacy in Vehicular Networks. Wiley-IEEE Press, (2015).
<https://ieeexplore.ieee.org/>

- [25] Banzi, M., & Shiloh, M. (2014). Getting Started with Arduino The Open-Source Electronics Prototyping Platform. O'Reilly Media.
https://link.springer.com/chapter/10.1007/978-3-642-59742-8_54
- [26] Monk, S. (2013). Programming Arduino Getting Started with Sketches. McGraw-Hill.
https://link.springer.com/chapter/10.1007/978-3-642-59742-8_54
- [27] Blum, J. (2013). Exploring Arduino Tools and Techniques for Engineering Wizardry. Wiley.
https://link.springer.com/chapter/10.1007/978-3-642-59742-8_54
- [28] Margolis, M. (2011). Arduino Cookbook. O'Reilly Media.
https://link.springer.com/chapter/10.1007/978-3-642-59742-8_54
- [29] Date, C. J. (2004). An Introduction to Database Systems. 8th ed. Addison-Wesley.
https://link.springer.com/chapter/10.1007/978-3-642-59742-8_54
- [30] Elmasri, R., & Navathe, S. B. (2015). Fundamentals of Database Systems (7th ed.). Pearson Education.
https://link.springer.com/chapter/10.1007/978-3-642-59742-8_54
- [31] Robinson, S. P., & Judge, T. A. (2019). Principles of Management.
https://link.springer.com/chapter/10.1007/978-3-642-59742-8_54
- [32] Kurose, J. F., & Ross, K. W. (2016). Computer Networking: A Top-Down Approach. Pearson.
https://link.springer.com/chapter/10.1007/978-3-642-59742-8_54
- [33] Berners-Lee, T., Fischetti, M., & Foreword By-Dertouzos, M. L. (2001). Weaving the Web: The Original Design and Ultimate Destiny of the World Wide Web by Its Inventor. Harper San Francisco.
https://link.springer.com/chapter/10.1007/978-3-642-59742-8_54
- [34] Pressman, R. S. (2014). Software Engineering A Practitioner's Approach. McGraw-Hill Education.
https://link.springer.com/chapter/10.1007/978-3-642-59742-8_54

- [35] Wright, S.P., Hall Brown, T.S., Collier, S.R., & Sandberg, K. (2017). How consumer physical activity monitors could transform human physiology research.
<https://doi.org/10.1152/ajpregu.00003.2017>
- [36] Piwek, L., Ellis, D.A., Andrews, S., & Joinson, A. (2016) The Rise of Consumer Health Wearables: Promises and Barriers.
<https://doi.org/10.1371/journal.pmed.1001953>
- [37] Evenson, K.R., Goto, M.M., & Furberg, R.D. (2015) "Systematic review of the validity and reliability of consumer-wearable activity trackers.
<https://ijbnpa.biomedcentral.com/articles/10.1186/s12966-015-0314-1>
- [38] Zhao, Y., et al. (2018). "Wearable Health Monitoring Devices: A Review of Technology, Application, and Future Directions". IEEE Access (pages 8-1).
<https://ieeexplore.ieee.org/document/8405184>
- [39] Pina, L. M., & Capobianco, M. (2020). "The Impact of Wearable Fitness Trackers on Health: A Systematic Review". Journal of Medical Internet Research (pages 12-1).
<https://www.jmir.org/2020/8/e17377>
- [40] Rao, A., & Patel, S. (2019). "Emergency Alert Systems and Wearable Technology: Current Trends and Future Directions". Health Information Science and Systems. (pages 8-1).
<https://hissjournal.biomedcentral.com/articles/10.1186/s13755-019-0303-0>
- [41] Kim, K., et al. (2020). "Integration of Wearable Health Monitoring Devices with Mobile Health Applications". Journal of Biomedical (pages 14-1).
<https://www.sciencedirect.com/science/article/pii/S1532046420300461>
- [42] Zhu, H., et al. (2019). "Wearable Sensors and Systems for Health Monitoring: A Review". Sensors. (pages 24-1).
<https://www.mdpi.com/1424-8220/19/23/5128>

- [43] Bong, C., et al. (2018). "Adherence Monitoring and Reminder Systems for Medication A Review". Biomedical Engineering Reviews.
- <https://link.springer.com/article/10.1007/s12195-018-0563-3>
- [44] Huang, C., et al. (2019). "Stress Monitoring and Management Using Wearable Technology". Frontiers in Psychology. (pages -1 12).
- <https://www.frontiersin.org/articles/10.3389/fpsyg.2019.02589/full>
- [45] Schlingensiepen, J., et al. (2015). Empowering People with Disabilities Using Urban Public Transport. Procedia Manufacturing, 3, 2358-2365.
- <https://linkinghub.elsevier.com/retrieve/pii/S2351978915004746>
- [46] Alam, T., et al. (2021) Designing a People Tracking System for Smart Cities. International Journal of System Assurance Engineering and Management, 12(1), (pages 1-9).
- <https://link.springer.com/article/10.1007/s13198-021-01072-4>
- [47] Bolsunovskaya, M., et al. (2022) Innovative Project Model Development for Wearable Devices.
- <https://link.springer.com/>
- [48] Raghavan, et al. (2020) Child Tracking System Using Android Phones.
- <https://link.springer.com/article/10.1007/s13198-020-00958-z>